

Current & Circuits

One chain: potential difference supplies energy per charge → charge moves → current flows → resistance opposes the flow.

CORE DEFINITIONS

Current

- Electric current = rate of flow of charge.
- $I = \Delta q / \Delta t$.
- Conventional current is + → −; electron flow is opposite.

Potential difference

- Work done per unit charge.
- $V = W/q$.
- $1 \text{ V} = 1 \text{ J/C}$.
- For an electron accelerated through V: $E = qV$; $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$.

OHM · RESISTIVITY · POWER

$$V = IR \cdot R = V/I$$

$$R = \rho L/A$$

$$P = IV = I^2 R = V^2/R$$

Remember: R depends on component conditions; ρ is a property of the material. Longer wire → higher R. Larger cross-sectional area → lower R.

MEASURING CIRCUITS

Ammeter

- Measures current.
- Connect in series.
- Ideal ammeter: $R = 0$.

Voltmeter

- Measures p.d. across a component.
- Connect in parallel.
- Ideal voltmeter: $R = \infty$.

SERIES vs PARALLEL

Rule	Series	Parallel
Current	Same everywhere	Splits: $I = I_1 + I_2 + \dots$
Voltage	$V = V_1 + V_2 + \dots$	Same across branches
Resistance	$R = R_1 + R_2 + \dots$	$1/R = 1/R_1 + 1/R_2 + \dots$
Quick check	Total R > any single R	Total R < smallest branch R

REAL BATTERIES

EMF & internal resistance

- EMF ε = total energy supplied per charge.
- Real battery has internal resistance r .
- $\varepsilon = I(R + r)$.

Terminal voltage

- $V = \varepsilon - Ir$.
- Ir = 'lost volts' inside the battery.
- As current increases, terminal voltage falls.

SENSORS & POTENTIOMETER

NTC thermistor / LDR

- NTC: $T \uparrow \rightarrow R \downarrow$.
- LDR: light intensity $\uparrow \rightarrow R \downarrow$.
- Both are non-ohmic environmental sensors.

Potentiometer

- Three-terminal variable resistor with a sliding wiper.
- Acts as a variable voltage divider.
- Output can vary from 0 up to the supply p.d.

TOP EXAM TRAPS

Fast checks

- Conventional current direction is not electron-flow direction.
- Voltmeter → parallel; ammeter → series.
- Adding a resistor in parallel decreases total resistance.
- Double wire length with same material/area → double R.
- Use $V = IR$ only where the component behaves as an ohmic conductor under the stated conditions.

STUDY METHOD: cover the right side / formula boxes → recall from memory → check → repeat.